

INTRODUCTION TO TIME SERIES ANALYSIS & FORECASTING :

A Python Implementation

II. Decomposition Methodology

The first approach to determining the presence of the various components in a time series is naturally by proceeding to the examination of the graphical representation of the series. Each component is hence expected to have a specific influence of the graphical aspect of the series. But this depth of analysis will not suffice in accurately providing answers to our various questions.

B. Detection, Extracting and Forecasting the Trend

The analysis procedure of the trend component is identical to that of the seasonal component. We'll be using the sensibly the same tools and drawing the appropriate conclusions

1. Component Detection methods
 - a. Graphical examination
 - b. Tabular analysis : the BB approach
 - c. Variance analysis test : the Row effects tests

Periodic influence (row effect) test:

$$H_0 : \text{No influence}$$

Compute $F_c = \frac{V_p}{V_R}$ and compare it to a $\mathcal{F}_{v_1; v_2}$ Fisher value where $v_1 = p - 1$ and $v_2 = (P - 1) * (S - 1)$

If $F_c > \mathcal{F}_{v_1; v_2}$ then H_0 is rejected => presence of trend Component

2. Detrending Procedure

Once it has been established that a trend is present in the series, the next step is to construct said trend such that we can compute forecast values, then remove the trend from the series (= detrend).

There are 2 main approaches to approximate a trend:

- a. Model based Detrending: A Regression approach

When it's been established that a trend component is present in the series, the correct use of the techniques presupposes that one can distinguish between deterministic trends and random trends. The deterministic trend is that which has an analytical or functional form. It can be linear or Non-linear.

In the case of a linear trend, the corresponding model is written: $X_t = \alpha + \beta t + \varepsilon_t$. The estimates and test results of this model can easily be obtained using OLS methods.

Then in case of a non-linear trend but that can undergo linear transformation (power function, logistic shape, exponential ...), the idea would be to conduct OLS of the linearized data.

The last case is that of a non-linear and non-transformable trend, in which case, the best line of action would be to opt for nonlinear modeling and or non-linear optimization algorithms.

b. Detrending via filter: Moving Averages and Smoothing

In the case of a time series affected by a random trend, *i.e.* when its evolution cannot be modeled by a functional form, we will resort to filtering methods, notably smoothing methods. Filtering methods are mathematical operations consisting in transforming an input X_t into a new output series Y_t .

The simplest forms of filtering methods are the moving average filters. The moving average of period length p , noted $MA_p(X_t)$, is a time series obtained by calculating the average of p values from the observed series X_t .

- It is an average because it is computed at a given time t , by taking the average of p values of the examined series *i.e.* $MA_p(X_t) = \frac{1}{p} \sum_{i=1}^p X_{t-i}$
- It is moving because it is calculated at each point in time throughout the history of the values.

In its most general conception, when considering a window size p that can be written $p = l + r + 1$, the associated moving average can be written: $MA_p(X_t) = \sum_{h=-l}^r \omega_h X_{t+h}$, where h is a time horizon centered around t and varying between $(t - l)$ and $(t + r)$, and ω_h is an assigned weight such as $\sum_{h=-l}^r \omega_h = 1$.

4 cases to distinguish:

- When $r = 0$, this implies that only historical and contemporaneous values of the series are used to compute the moving average.
- When using moving averages for forecasting purposes, only historical values of the series are used.
- When $r \neq 0$, this indicates for any given time t , future values of the series are considered in the computation of the moving average. This is what we will refer to as smoothing a series. Plus, when $l = r$, the moving average is said to be centered.
- The most commonly used method for computing a centered MA consists in taking $p+1$ values and dividing by 2 the first and last of these values.

A bit more advanced smoothing methods that we shall be using are Exponential Smoothing methods. These methods consist of empirical techniques that have the common characteristics of giving greater weight to contemporary values of the series in the smoothing process. In the absence of a seasonal component, they are structured as follows:

$$X_t = f_t^{(k)} + \varepsilon_t$$

- where ε_t is stationary random process (*i.e.* $E[\varepsilon_t] = 0$; $E[\varepsilon_t^2] = \sigma_\varepsilon^2$; $E[\varepsilon_t \varepsilon_{t'}] = 0$)
- $f_t^{(k)}$ is a polynomial function of order k with time-varying parameters

Smoothing methods differ between each other, based on the order k of the $f_t^{(k)}$ function, the most common values for k being $k = 0$ and $k = 1$.

- For $k = 0$, the data has the following structure:

$$X_t = a_t + \varepsilon_t \quad \forall t$$

with $a_t = a$ constant on local intervals of the series. On this type of data structure, the most adapted method is the simple exponential smoothing method.

- For $k = 1$, the we have:

$$X_t = a_t + b_t * t + \varepsilon_t$$

With $a_t = a$ and $b_t = b$ on considered local intervals. For such structure, we'll be using the Double exponential smoothing method.

Cf appendices for more detailed presentation of Exponential smoothing methods.

Class exercises shall provide more in-depth understanding of these notions.

Simple Exponential Smoothing - SES

The idea behind SES is that a smoothed version of a series can be obtained by computing each value as a combination of historical values with decreasing weights. In other words:

$$z_{t-1} = b_1 X_{t-1} + b_2 X_{t-2} + \dots + b_k X_{t-k} \text{ with } b_1 > b_2 > \dots > b_k$$

The z_{t-1} value is the output of the exponential smoothing filter, computed at time $t - 1$. With the b_k parameters estimated as previously explained, this value can also be seen as the forecast of X_t , computed at $t - 1$. $\Rightarrow z_{t-1} = X_{t-1,t}^f = \hat{X}_t$

The only issue left is to obtain the b_k parameters such that they decrease exponentially (= geometric progression), and they sum up to 1 (= normalized).

$$b_k = b(1 - \lambda)^k \quad 0 \leq \lambda \leq 1 \quad k = 1, 2, \dots \text{ and } b_k \in \mathbb{R}$$

The normalized weights are obtained as: $\omega_k = \lambda(1 - \lambda)^{k-1} \Rightarrow \sum_{k=1}^{\infty} \omega_k = 1$

Replacing these weights in the geometric progression gives:

$$\begin{aligned} z_{t-1} = \hat{X}_t &= \lambda X_{t-1} + \lambda(1 - \lambda)X_{t-2} + \lambda(1 - \lambda)^2 X_{t-3} + \lambda(1 - \lambda)^3 X_{t-4} + \dots \\ &= \lambda \sum_{i=1}^{\infty} (1 - \lambda)^{i-1} X_{t-i} \end{aligned}$$

And finally, be rewritten: $\hat{X}_t = \lambda X_{t-1} + (1 - \lambda)\hat{X}_{t-1} \Rightarrow$ the simple exponential smoothing is the result of a weighed average between the last observation and the last smoothed value.

The λ parameter is called the smoothing constant and represents the degree of memory in the series.

- Values of λ close to 0 indicated that the smoothing weighing process incorporates a large number of historical values. The forecast is not very sensitive to the latest observations.
- Values of λ close to 1 indicated that contemporaneous observations have a preponderant weight over the old terms, the memory of the phenomenon is weak, and the smoothing is very reactive to the last observations.

Double Exponential Smoothing - DES

The concept of Double exponential smoothing comes in handy when we are dealing with a series containing a random trend. As this trend component cannot be linearized through some transformation, the idea here is to consider that while the trend is globally nonlinear, it can, however, be seen as locally linear. The objective hence being to model a trend component dans is adaptive in nature to account for the changes in intercept and slope of the trend, in its various considered regions.

AS the name indicates, Holt's double exponential smoothing consists in smoothing an already smoothed version of a series that can be described as follows:

$$X_t = a_t * t + b_t + \varepsilon_t \text{ with } a_t = a \text{ and } b_t = b \quad \forall t, t \in [n_1; n_2] \subset [1; n]$$

The steps for computing the double exponential smoothing are as follows:

1. Smoothing the initial series as done via SES

$$\hat{X}_t = \lambda X_{t-1} + (1 - \lambda)\hat{X}_{t-1}$$

2. Smoothing is resulting series in the same manner (ps: note that in this case smoothing is based on time t of the smoothed series)

$$\hat{X}_t = \lambda \hat{X}_{t-1} + (1 - \lambda) \hat{X}_{t-1}$$

3. Computing the adaptive intercept values as: $b_t = 2\hat{X}_t - \hat{X}_t$
 4. Computing the adaptive slope values as: $a_t = \frac{1}{\lambda} (\hat{X}_t - \hat{X}_t)$ where $\bar{\lambda} = \frac{1-\lambda}{\lambda}$
- $$a_t = (\hat{X}_t - \hat{X}_t) * \frac{\lambda}{1 - \lambda}$$

5. Forecast values are then obtained as follows:
 - In sample forecasts: $F_t = a_t + b_t$, for $t \in [2, n + 1]$
 - Out of sample forecasts: $F_{t+h} = a_{n+1} * h + b_{n+1}$ for $h = 1, 2, \dots$

Hence, the forecast via DES is a line having intercept and slope values equal to the last computed values of a_t and b_t . We can extrapolate values on said line on a given h forecast horizon. It is however recommended to keep h as small as possible as the forecast values degrade rapidly as h increments.